Kaya Acarbay

Software Developer

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Menemen, İzmir, Türkiye

4th-year IT student building AI agents and autonomous systems. Intern at VeriTeknik, contributor to the open-source pluggedin-app, and founder of a Project Catalyst-funded Cardano education initiative.

Languages: Python, JavaScript, TypeScript, C++, HTML & CSS
Web & data: React, FastAPI, PostgreSQL
AI & tools: LLMs, AI agents, prompt engineering, Git

PROJECTS

Agent-Based Civilization Simulation · JavaScript, HTML5 Canvas — github.com/acarbayk/agent-based-civilization-simulation

- Built a seeded, deterministic browser simulation of up to 1,100 people in four kingdoms: the same seed replays the same history.
- Modeled each person with a Big Five personality, family, memories and goals using rule-based minds instead of per-person LLM calls.
- Tuned rebellion dynamics (Epstein 2002 civil violence model, threshold 0.13) and checked demography against historical ranges.

Bug-Hunter · Python — github.com/acarbayk/Bug-Hunter

- Built an autonomous bug bounty agent that automates vulnerability detection and CI/CD integration.

ChessBot · Python, python-chess — github.com/acarbayk/ChessBot

- Wrote a minimax chess engine with alpha-beta pruning (about 1200–1600 Elo) and a GUI that highlights moves.

EXPERIENCE

Support & Infrastructure Intern · VeriTeknik — Jul 2026 – Present

- Monitor infrastructure and resolve customer support tickets.
- Fixed a Streamable HTTP transport key propagation bug (#172) in pluggedin-app, an open-source MCP server platform (TypeScript).

Founder · Spreading Cardano in Turkiye — Nov 2023 – May 2024

- Won 40,333 ADA of Project Catalyst Fund 10 funding and completed all three milestones.
- Produced podcast and YouTube content and organized community meetups to build a Turkish Cardano community from scratch.

Volunteer Mentor · Halit Özpirinç Anatolian High School, İzmir — Sep 2021 – Present

- Mentor high-school students on TÜBİTAK research projects.
- Led Apollo 6, a circular farming project submitted to Teknofest.

Co-Founder & Activity Manager · EMU RPG Club, Famagusta — Nov 2023 – Mar 2025

- Grew the club to nearly 300 members and secured two company sponsorships.
- Ran weekly sessions and tournaments; co-developed the club website.

EDUCATION

B.Sc. Information Technology · Eastern Mediterranean University, Famagusta — Expected June 2027

High School Diploma · Halit Özpirinç Anatolian High School, İzmir · Erasmus+ exchange, Portugal — 2021

Certificates: AWS Training & Certification: 8 completions (6 machine learning, 2 generative AI), Jul 2025 · Udemy: Python Beginner to Advanced (40+ h), Feb 2023
Languages: Turkish (native) · English (advanced) · Japanese (beginner)